

Analysis Report

1. Introduction

Project objective:

Demonstrate electrical circuit analysis using Python.

2. Ohm's Law Analysis

Input:

Voltage = 12 V

Resistance = 6 Ω

Calculation:

Current = Voltage / Resistance

Result:

Current = 2 A

3. KCL Verification

Node Current Analysis

Incoming currents:

$I_1 = 2 \text{ A}$

$I_2 = 3 \text{ A}$

Outgoing currents:

$I_3 = 5 \text{ A}$

Verification:

$I_1 + I_2 = I_3$

$$5 \text{ A} = 5 \text{ A}$$

KCL satisfied.

4. KVL Verification

Loop Voltage Analysis

Source Voltage:

$$12 \text{ V}$$

Voltage Drops:

$$R1 = 4 \text{ V}$$

$$R2 = 8 \text{ V}$$

Verification:

$$12 - 4 - 8 = 0$$

KVL satisfied.

5. Conclusion

The Python implementation successfully validates:

- Ohm's Law
- Kirchhoff's Current Law
- Kirchhoff's Voltage Law